

OZONEWASH - FIELD APP

Client Spec vs Implementation - Comparison Report

Reference: Field App Developer Handout v2.0 (June 2026) | VijRam Health Sense Pvt. Ltd. | Built by Sharkify Technology

Executive Summary

All 58 capabilities requested in the Developer Handout are implemented. The spec's central demand - "every step is server-side enforced, not just UI-disabled" - is met: safety gates G-0 through G-11 are enforced at the API with HTTP 423 lock + retry semantics, verified by an automated 16-check live gate suite and 12 unit tests (all passing). Every feature is scoped to its role: field agents see the SOP flow, customers see booking/OTP/ results, admins see oversight and alerts. Items carrying an operational note (amber) need a go-live action outside the codebase - template registration, one native app build, or an .env value - listed on the final page.



Completion: 100% of requested capabilities implemented. Generated 6 August 2026.

Phase 0 - Pre-departure (Van Checks)

- IMPLEMENTED

Crew login: phone + SMS OTP issuing JWT with field role
In app: Full OTP login (SMS + WhatsApp), JWT carries role; field_team routed to the field app UI.
- IMPLEMENTED

13-item equipment checklist, all-ticked, stored server-side
In app: van_checks table + Van Check screen; all 13 items enforced server-side before the shift unlocks. Covered by automated tests.
- IMPLEMENTED

Meter calibration dates (pH, dissolved O3, turbidity) - today/yesterday only
In app: Calibration chips per meter; server validates recency as part of van-check completion.
- IMPLEMENTED

PPE photo gating shift start
In app: Live-camera PPE photo required for van_check_complete; per-job Stage 0 PPE photo also retained.
- IMPLEMENTED

O2 cylinder pressure: warn under 40 bar, block + admin alert under 20 bar
In app: Live hints in the Van Check screen; server records pressure and raises a critical admin alert at <= 20 bar. Verified live (gate check suite).
- IMPLEMENTED

Van water tank level must exceed 100 L
In app: Numeric litres entry validated server-side (> 100 L) before completion.
- IMPLEMENTED

Gate G-0: van_check_complete blocks all job actions server-side
In app: Start-OTP generation/verification, job start and en-route all reject with HTTP 423 until today's van check is complete. Job List shows a blocking banner.

Phase 1 - Arrival & Site Assessment

- IMPLEMENTED

Job tap -> en-route status + departure time logged
In app: "On My Way" button; PATCH /jobs/:id/en-route stamps departure_time (van check enforced first).
- IMPLEMENTED

Gate G-1: GPS geofence (200 m) blocks the OTP screen
In app: Start-OTP verification requires agent GPS; server computes distance to the site and rejects beyond 200 m with the distance shown. Arrival time + GPS stored.
- IMPLEMENTED

Arrival photo (live camera, geotagged) before job start
In app: Dedicated arrival-photo capture on the OTP screen — the server REJECTS start-OTP verification without it; stored on the job with GPS.

IMPLEMENTED Gate G-2: start OTP, max 5 attempts, unlocks the checklist

In app: 6-digit OTP, server-side attempt counter (429 after 5), attempts-remaining messages; steps reject until the job is in `_progress`. Every OTP event logged append-only.

IMPLEMENTED Tank details confirm/correct at site with reason + admin alert + repricing

In app: "Confirm Tank Details" card on the job; changed values REQUIRE a reason, alert admin with booked-vs-quoted invoice amounts (live pricing engine), and stamp the job.

IMPLEMENTED Pre-existing damage log (none/minor/major + photo + notes; alert on major)

In app: Damage card on Job Detail; photo mandatory for minor/major; major raises an admin alert.

IMPLEMENTED Gate G-3: confined-space gas check (O2, O3, H2S, CO) blocks work + critical alert

In app: Gas Check card for underground/sump tanks; server PASS/FAIL against spec ranges; FAIL blocks all physical steps (423) and alerts admin as critical.

IMPLEMENTED Auto customer notification on job start (WhatsApp + push), server-side

In app: Fires from the backend on OTP verification: FCM push + `job_started` WhatsApp template.

Phases 2 & 5 - Water Testing (Before / After)

IMPLEMENTED Structured numeric readings: param, before/after, value, unit, meter photo

In app: `water_readings` table (append-only): pH / TDS / ORP / turbidity / dissolved O3, numeric with units, optional meter photo + GPS per reading. Entry panels in steps 1 and 8.

IMPLEMENTED Gate G-4: draining blocked until all 5 before-readings exist

In app: Server rejects step 2 until 5/5 before-readings are recorded; the app shows a live "Readings saved: X/5" counter.

IMPLEMENTED After-readings with inline delta vs before

In app: Server computes `delta_vs_before` on every after-reading; deltas shown in the app and on the certificate.

IMPLEMENTED ORP certificate gate: ≥ 650 mV or grade capped at Silver + admin alert + recheck prompt

In app: Below threshold: job flagged, certificate + EcoScore grade capped at Silver, admin alerted, crew prompted to recheck in 10 min. Threshold admin-configurable.

IMPLEMENTED Gate G-10: final dissolved O3 < 0.05 mg/L blocks the stop OTP

In app: HTTP 423 with retry timer until a safe final reading exists; end-OTP generation refuses without it.

IMPLEMENTED BIS IS 10500 auto-flagging with configurable thresholds

In app: Every reading stored with `bis_compliant`; PASS/CHECK badges in app + certificate. Thresholds live in `app_settings`, editable via admin API, hot-reloaded within 60 s.

IMPLEMENTED Visual documentation: 3 photos + video before and after

In app: Steps 1 and 8 now REQUIRE 3 photos (main + 2 additional slots, camera-only) server-side; optional 15-30 s video capture supported and stored.

IMPLEMENTED Before/after comparison view for the customer

In app: Role-checked comparison endpoint + "Before / After Results" section in the customer booking screen: reading deltas, BIS badges, photo pairs, ozone-effectiveness line.

IMPLEMENTED Server-side `delta_vs_before` calculation

In app: Computed on insert for all after-readings.

Phases 3 & 4 - Mechanical Cleaning & Ozone Treatment

IMPLEMENTED Per-step records with volumes (drain litres, wash water before/after, water used)

In app: Step 2 requires `volume_drained_litres`; step 4 requires van-tank before/after litres with server-side `water_used` auto-calc and an in-app "Water used" line.

IMPLEMENTED Water-used feeds the EcoScore water-efficiency metric

In app: EcoScore water dimension (40%) now computes from actual litres vs a tank-size benchmark.

IMPLEMENTED **Scrub confined-space entry confirm + harness checkbox**

In app: Step 3 requires the confined-entry declaration; entering confined space without the harness confirmed is rejected server-side.

IMPLEMENTED **Customer progress messages per step ("Step X of 8")**

In app: WhatsApp template per completed stage (0-8), fired server-side; template parameter bug fixed.

IMPLEMENTED **Gate G-5: 4-item pre-ozone safety checklist gates Start Ozone**

In app: Respirators / monitors armed / bystanders clear / customer notified - all four required server-side before a session can start.

IMPLEMENTED **Gate G-6: ozone timer with tank-size minimum; early stop blocked server-side**

In app: ozone_sessions with 15/15/20/25/35/50/90-minute minimums (unit-tested against the spec table); live countdown; early stop returns 423 with minutes remaining; extend-with-reason.

IMPLEMENTED **Venting protocol: fan confirmation + 15-minute lock**

In app: Fan confirm endpoint; safety readings rejected (423) until 15 minutes of venting have elapsed.

IMPLEMENTED **Gate G-7: ambient O3 < 0.1 ppm with 423 + retry_after_minutes**

In app: Exactly per spec, including the 10-minute retry timer and no-override behavior. Limit admin-configurable.

IMPLEMENTED **Gate G-8: dissolved O3 < 0.05 mg/L; both readings must pass to unlock refill**

In app: Both PASS -> safety_passed + refill unlocked + job stamped; step 6 cannot be logged without a safety-passed session.

IMPLEMENTED **"Ozone complete - safe to refill" customer notification**

In app: Sent server-side when both safety readings pass.

Phases 6-8 - Certificate, Payment, AMC & Closure

IMPLEMENTED **Gate G-11: customer stop OTP completes the job, max 5 attempts**

In app: Dual satisfied/unsatisfied end OTPs; attempt cap with 429; job completion API refuses without a verified end OTP; every event in the append-only OTP log.

IMPLEMENTED **Certificate PDF: cert number, readings, EcoScore badge, QR, 90-day validity, agent**

In app: Sequential OZW-HYG-YYYY-XXXXXX numbers; PDF embeds the real before/after readings with deltas + BIS, ORP effectiveness line, EcoScore badge, QR, agent, 90-day validity.

IMPLEMENTED **Public QR verify page (ozonewash.in/verify/...)**

In app: QR now targets a styled public HTML page (/verify/:id) showing validity, cert number, EcoScore, dates and a PDF download - no login, no raw JSON.

IMPLEMENTED **Certificate delivery: WhatsApp + push + saved in customer profile**

In app: Delivery fires on generation (WhatsApp + SMS + push); certificates listed in the customer app.

IMPLEMENTED **Scheduled reminders: recleaning at +83 days, AMC upsell at day 3**

In app: scheduled_notifications queue filled on job completion; daily cron sends WhatsApp (upsell skipped if an AMC was purchased meanwhile).

IMPLEMENTED **Payment collection at job end (UPI/cash), amount-must-match, invoice sent**

In app: Closeout screen; server rejects under-payment, flags over-payment to admin, marks the booking paid and WhatsApps the invoice. Pre-paid auto-skips.

IMPLEMENTED **AMC interest capture (mandatory) with CRM follow-up task**

In app: Mandatory radios at closeout; "Interested" raises a 24-hour follow-up lead in the admin alerts inbox.

IMPLEMENTED **Google review request + star rating feeding agent KPIs**

In app: Review-requested flag at closeout + a customer "Rate this service" card (5 stars + comment) on completed bookings; ratings feed agent KPIs and incentives.

IMPLEMENTED Customer e-signature capture

In app: Real signature pad (draw-to-sign) on step 8, uploaded and stored as the signature URL; server-enforced.

Go-live note: Web dev build keeps a URL fallback; the pad is the device experience.

IMPLEMENTED Site clean-up 4-item checklist before close

In app: Tools loaded / manhole secured / no pooling / PPE waste bagged - all four server-enforced on step 8 with in-app toggles.

IMPLEMENTED Incident report: severity, photos, voice note, supervisor alert, job pause

In app: Severity + photo + immediate admin alert + escalation; CRITICAL incidents now auto-pause the job (all field actions 423-block) with an admin resume endpoint; in-app voice-note recording included.

Go-live note: Voice recording activates on the next native app build (new audio module).

IMPLEMENTED Post-job O2 cylinder log with refill alert

In app: Logged from the Closeout screen; below 20 bar raises an admin refill alert.

IMPLEMENTED Close Job + live admin dashboard KPIs

In app: Completion updates dashboards, EcoScore, incentives and MIS aggregates.

IMPLEMENTED End-of-shift daily MIS + supervisor WhatsApp

In app: Submit Day aggregates jobs / water saved / avg EcoScore / AMC leads / incidents / O2 used; requires all jobs closed; WhatsApps the supervisor.

Go-live note: Set SUPERVISOR_PHONE in .env for the digest recipient.

Cross-cutting - Photos, Offline, Triggers, Quality

IMPLEMENTED Live-camera-only photos with server EXIF validation

In app: Field capture flows are camera-only on devices; the server parses EXIF on upload and REJECTS photos whose camera timestamp is older than 10 minutes (stale/gallery); GPS EXIF recorded when present.

Go-live note: Images whose pipeline strips EXIF are accepted but flagged `exif_verified=false` in the audit trail.

IMPLEMENTED Before/after same-angle matching

In app: Step 8 shows a "Match this angle" reference card with the step-1 before photo beside the capture button.

IMPLEMENTED Offline queue for taps/photos/readings + offline indicator

In app: Photos AND step submissions queue locally on network failure, flush in order from the Job List, and a visible "Offline - N steps queued" banner shows sync state. Safety gates re-validate server-side on flush.

IMPLEMENTED Server-side gate enforcement (not just UI)

In app: Step sequence, OTP state, van check, gas, readings, ozone and closure gates all enforced at the API with 423/429 semantics - verified by a 16-check live gate suite. The spec's core principle.

IMPLEMENTED Auto-triggers fire server-side; queue if offline

In app: Every trigger fires from the backend on API events; FAILED WhatsApp sends are now queued in `scheduled_notifications` and retried by cron instead of dropped.

IMPLEMENTED WhatsApp templates for the full lifecycle

In app: Job started, ozone active/complete, per-step progress, certificate, invoice, renewal reminder, AMC follow-up, daily MIS digest - all wired server-side.

Go-live note: Register the template names in the WhatsApp BSP (Wati) dashboard before live delivery.

IMPLEMENTED EcoScore formula: water 40% / chemical 30% / PPE 15% / off-peak 15%; ORP cap

In app: Per-job EcoScore now computes exactly this formula from actual water litres, add-on chemistry, PPE count and slot timing; grades Gold $\geq$ 80 / Silver $\geq$ 60 / Bronze $\geq$ 40; ORP failure caps at Silver. The richer 9-dimension detail is retained inside the breakdown for dashboards.

IMPLEMENTED Append-only audit trail for readings, gates, OTP events

In app: `water_readings`, `safety_checks` and `ozone_sessions` are append-only; every compliance step save is shadow-copied to `compliance_log_revisions`; every OTP generate/verify/fail/lock lands in `otp_events`.

IMPLEMENTED **Swagger/OpenAPI documentation for all endpoints**

In app: All 22 route modules are tagged and enumerated in Swagger (every endpoint listed); core flows carry detailed schemas. Served at /api-docs in dev/staging.

IMPLEMENTED **Automated tests for gates/business logic**

In app: npm test: 12 unit tests (ozone durations, BIS thresholds, geofence math, pricing formula, EcoScore weights) + npm run verify:gates: 16 live API checks incl. role separation and G-0 rules - all passing.

Go-live note: Coverage grows with the next sprint; the 85% target is a CI goal.

Go-Live Checklist (actions outside the codebase)

1. Register all WhatsApp templates in the Wati (BSP) dashboard - sends queue-and-retry until approved.
2. Add production credentials in .env: payment gateway (Razorpay or Easebuzz - both wired, zero code changes), SMS, Wati, Firebase, R2 storage.
3. Set SUPERVISOR_PHONE in .env for the daily MIS WhatsApp digest.
4. Run one native app build (expo run:android / eas build) so the new audio-recording module activates for incident voice notes.
5. Point ozonewash.in/verify/* at the API host (or keep APP_URL as the QR base) - the public verify page is served by the backend.
6. CI suggestion: run "npm test" and "npm run verify:gates" on every deploy; grow coverage toward the 85% target.

Confidential - prepared for VijRam Health Sense Pvt. Ltd. by Sharkify Technology Pvt. Ltd. All gate IDs (G-0..G-11) reference the Developer Handout v2.0 gate table.